Satellite

A satellite is an object that moves around a larger body in space, following a fixed path called an orbit, and it can be either natural, like the Moon orbiting Earth, or artificial, such as human-made machines launched into space to perform specific tasks; these artificial satellites are used for many important purposes, including communication, weather forecasting, navigation (like GPS), and scientific research, as they can observe large areas of Earth and space from above the atmosphere, sending data back using radio signals, and they remain in orbit because their forward motion balances the pull of gravity, preventing them from falling back to Earth or drifting away into space.

The IM-1 Lander

The IM-1 lander, also known as “Odysseus,” was a robotic lunar mission developed by the company Intuitive Machines and launched in February 2024 as part of NASA’s Commercial Lunar Payload Services program. It travelled to the Moon aboard a SpaceX Falcon 9 rocket and successfully landed near the Moon’s south polar region on February 22, marking the first-ever soft landing by a private company and the first American lunar landing since 1972. The lander carried several scientific instruments, including six NASA payloads, designed to study the lunar surface, test new landing technologies, and gather data to support future human missions under the Artemis program. Although the lander tipped slightly upon landing, it remained operational and transmitted valuable data for about a week before shutting down as the lunar night began, making the IM-1 mission a major milestone in commercial space exploration.